

OSSP

Practical Week -1

Roll No – 2520090085

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Task 1-:

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

- a) Write a C code to create a process using fork() system call.
- b) Read the user input as shell command.
- c) Use a system call exec() to execute the shell command.
- d) Use a wait() system call for waiting a parent process.
- e) Display the PIDs of parent and child processes.

```
GNU nano 8.7.1                                     command_runner.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

#define MAX_CMD_LEN 1024
#define MAX_ARGS 64

int main() {
    char input[MAX_CMD_LEN];
    char *args[MAX_ARGS];

    // Read user input as shell command
    printf("Enter Linux command: ");
    if (fgets(input, sizeof(input), stdin) == NULL) {
        perror("Error reading input");
        exit(EXIT_FAILURE);
    }

    // Strip newline character
    input[strcspn(input, "\n")] = '\0';

    if (strlen(input) == 0) {
        printf("No command entered.\n");
        return 0;
    }

    // Parse command arguments
    int i = 0;
    char *token = strtok(input, " ");
    while (token != NULL && i < MAX_ARGS - 1) {
        args[i++] = token;
        token = strtok(NULL, " ");
    }
    args[i] = NULL;

    // Display Parent Process PID
    printf("\n[Parent Process] PID: %d\n", getpid());
```

```

// Display Parent Process PID
printf("\n[Parent Process] PID: %d\n", getpid());

// Create child process using fork()
pid_t pid = fork();

if (pid < 0) {
    perror("Fork failed");
    exit(EXIT_FAILURE);
}
else if (pid == 0) {
    // Child Process
    printf("[Child Process] PID: %d (Parent PID: %d)\n", getpid(), getppid());
    printf("[Child Process] Executing: '%s'...\n", args[0]);

    // Execute shell command using execvp()
    if (execvp(args[0], args) == -1) {
        perror("Execution failed");
        exit(EXIT_FAILURE);
    }
}
else {
    // Parent Process: wait for child
    int status;
    wait(&status);
    printf("\n[Parent Process] Child process (PID: %d) completed.\n", pid);
}

return 0;
}

```

```

nocturnal@nocturnal: ~/OSSP
Windows PowerShell
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PS C:\Users\vsbsu> wsl
nocturnal@nocturnal:/mnt/c/Users/vsbsu$ mv OSSP_2520090085 ~
mv: cannot stat 'OSSP_2520090085': No such file or directory
nocturnal@nocturnal:/mnt/c/Users/vsbsu$ cd ~/OSSP_2520090085
nocturnal@nocturnal:~/OSSP_2520090085$ ^C
nocturnal@nocturnal:~/OSSP_2520090085$ git init
Reinitialized existing Git repository in /home/nocturnal/OSSP_2520090085/.git/
nocturnal@nocturnal:~/OSSP_2520090085$ git branch Practical
fatal: a branch named 'Practical' already exists
nocturnal@nocturnal:~/OSSP_2520090085$ git switch Practical
Already on 'Practical'
nocturnal@nocturnal:~/OSSP_2520090085$ nano command_runner.c
nocturnal@nocturnal:~/OSSP_2520090085$ gcc command_runner.c -o command_runner
nocturnal@nocturnal:~/OSSP_2520090085$ ./command_runner
Enter Linux command: ls

[Parent Process] PID: 534
[Child Process] PID: 543 (Parent PID: 534)
[Child Process] Executing: 'ls'...

Practical_Week2 README.md command_runner command_runner.c

[Parent Process] Child process (PID: 543) completed.
nocturnal@nocturnal:~/OSSP_2520090085$ |

```

Task-2:

Using command `uname -a`:

```
nocturnal@nocturnal:~/OSSP_2520090085$ nano command_runner.c
nocturnal@nocturnal:~/OSSP_2520090085$ uname -a
Linux nocturnal 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
nocturnal@nocturnal:~/OSSP_2520090085$
```

Using command `lscpu`:

```
Linux nocturnal 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
nocturnal@nocturnal:~/OSSP_2520090085$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 39 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 16
On-line CPU(s) list: 0-15
Vendor ID: GenuineIntel
Model name: 13th Gen Intel(R) Core(TM) i5-13450HX
CPU family: 6
Model: 183
Thread(s) per core: 2
Core(s) per socket: 8
Socket(s): 1
Stepping: 1
BogoMIPS: 5222.39
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsc
cp_l1 constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pei pclmulqdq vmx ssse3 fma cx16 pcid
sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm sha3 dpmoprefetch ssbd sbrs i
bpb stibp lbrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx snap clflush
opt clwb sha_ni xsaveopt xsavec xsaveopt xsavec avx_vnni vmmi umip waitpgb gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_
clear serialize ibt flush_l1d arch_capabilities
Virtualization features:
Virtualization: VT-x
Hypervisor vendor: Microsoft
Virtualization type: full
Caches (sum of all):
L1d: 384 KiB (8 instances)
L1i: 256 KiB (8 instances)
L2: 16 MiB (8 instances)
L3: 20 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0-15
Vulnerabilities:
Gather data sampling: Not affected
Ghostwrite: Not affected
Indirect target selection: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Mitigation: Clear Register File
Retbleed: Mitigation: Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation: Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation: usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation: Enhanced / Automatic IBRS; IBPB conditional; PRRS0-eIBRS SW sequence; BHI BHI_DIS_S
Srbds: Not affected
Tsx: Not affected
Tsx async abort: Not affected
Vmscape: Not affected
nocturnal@nocturnal:~/OSSP_2520090085$
```

Using Command `lsblk`:

```
Vmscape: Not affected
nocturnal@nocturnal:~/OSSP_2520090085$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 3G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro
```

Using Command `ps`:

```
nocturnal@nocturnal:~/OSSP_2520090085$ ps
PID TTY TIME CMD
322 pts/0 00:00:00 bash
11589 pts/0 00:00:00 ps
nocturnal@nocturnal:~/OSSP_2520090085$
```

Using command top:

```
nocturnal@nocturnal:~/OSSP_2520090085$ top
top - 13:16:15 up 34 min, 1 user, load average: 0.10, 0.08, 0.08
Tasks: 27 total, 1 running, 26 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 11813.6 total, 11114.9 free, 627.1 used, 255.6 buff/cache
MiB Swap: 3072.0 total, 3072.0 free, 0.0 used, 11186.5 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24664   15832   11568 S   0.0   0.1   0:01.43 systemd
    2 root        20   0    3180    2208    2072 S   0.0   0.0   0:00.00 init-systemd(Ub
    6 root        20   0    3180    2044    1940 S   0.0   0.0   0:00.00 init
   43 root        19  -1   50424   16920   15608 S   0.0   0.1   0:00.16 systemd-journal
   78 systemd+    20   0   22540   14596   11976 S   0.0   0.1   0:00.03 systemd-resolve
   85 root        20   0   35456   12344    9156 S   0.0   0.1   0:00.56 systemd-udev
  103 root        20   0   28888   19488    1820 S   0.0   0.0   0:00.01 chronyd-starter
  104 root        20   0    4472    3136    2884 S   0.0   0.0   0:00.00 cron
  105 message+    20   0    9188    5772    4688 S   0.0   0.0   0:00.26 dbus-daemon
  157 root        20   0   44092   29800   14656 S   0.0   0.2   0:00.12 networkd-dispat
  161 root        20   0   18612    9500    8200 S   0.0   0.1   0:00.09 systemd-logind
  198 root        20   0    5492    2680    2500 S   0.0   0.0   0:00.00agetty
  207 _chrony      20   0   20420    6444    5668 S   0.0   0.1   0:00.06 chronyd
  216 _chrony      20   0   12092    2424    1784 S   0.0   0.0   0:00.00 chronyd
  228 root        20   0   123456   32556   17944 S   0.0   0.3   0:00.09 unattended-upgr
  320 root        20   0    3184    1112     980 S   0.0   0.0   0:00.00 SessionLeader
  321 root        20   0    3200    1248    1108 S   0.0   0.0   0:00.07 Relay(322)
  322 nocturn+    20   0    6268    5532    3860 S   0.0   0.0   0:00.09 bash
  323 root        20   0    8648    5532    4688 S   0.0   0.0   0:00.00 login
  365 nocturn+    20   0   22684   13664   11032 S   0.0   0.1   0:00.12 systemd
  367 nocturn+    20   0   22912    4112    2108 S   0.0   0.0   0:00.00 (sd-pam)
  394 nocturn+    20   0    6136    5412    3860 S   0.0   0.0   0:00.01 bash
   844 polkitd      20   0   380176   8588    7708 S   0.0   0.1   0:00.09 polkitd
  8776 syslog      20   0   220548   5508    4664 S   0.0   0.0   0:00.02 rsyslogd
  9529 nocturn+    20   0    8544    5092    4672 S   0.0   0.0   0:00.00 dbus-daemon
11518 root        20   0  1865776  14876   12140 S   0.0   0.1   0:00.04 wsl-pro-service
11603 nocturn+    20   0    8276    5680    3532 R   0.0   0.0   0:00.00 top
```